



SmartBimPei Suite — Technical Documentation & User Manual

Advanced Revit® Add-in Suite for Pre-Synthesis & Data
Compliance Control



Introduction & Core Value Proposition

Who is this tool for?

This plugin was designed by and for **BIM Managers, BIM Coordinators, Modelers** (HVAC, Plumbing, Electrical, Architecture), and **Company Executives**.

What is its main mission?

In the BIM world, clash detection and resolution (creating openings, communication) are time-consuming processes that

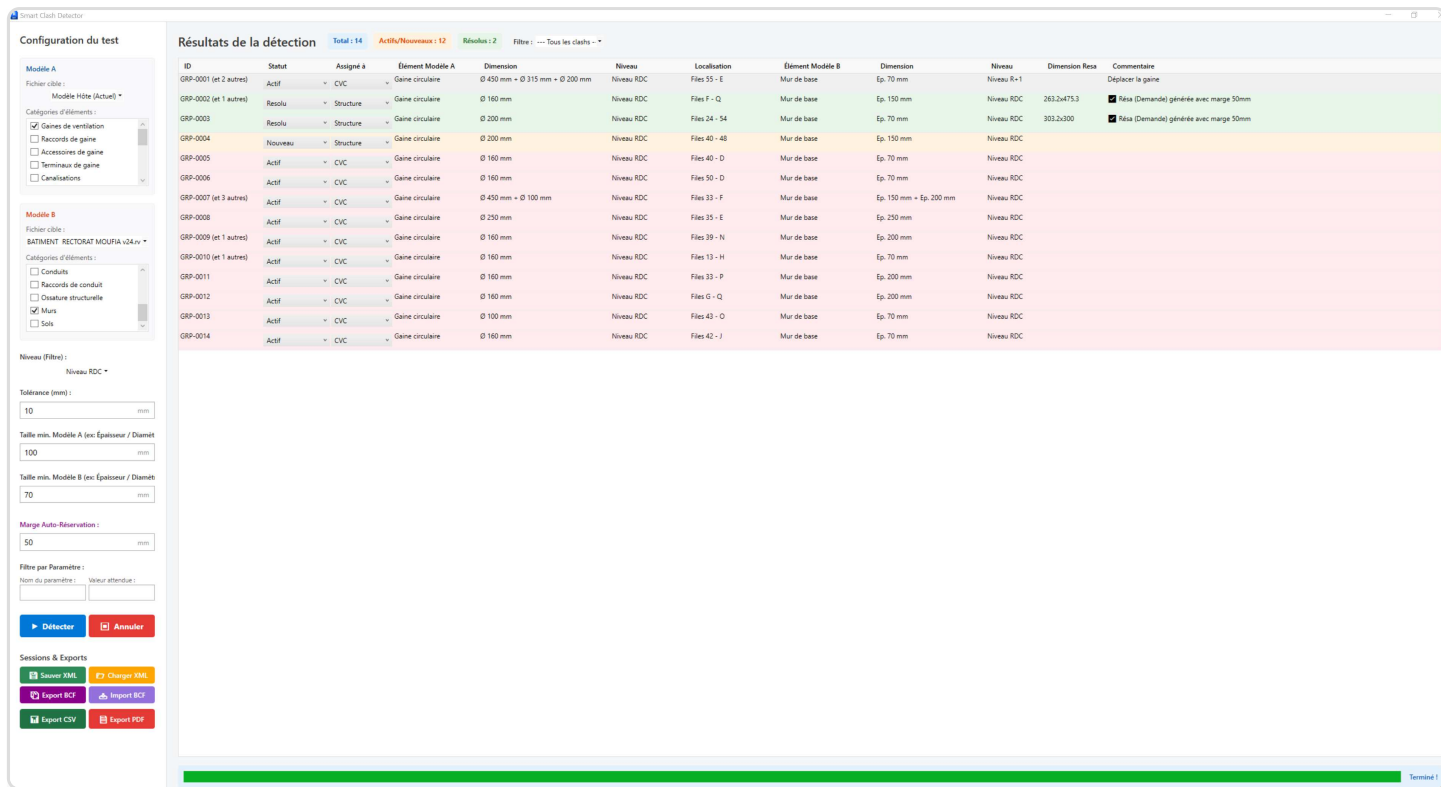
are extremely costly for companies. SmartBimPei eliminates these bottlenecks through:

- **Intelligent Automation:** Forget creating openings manually one by one. This plugin handles complex network fusions and curved walls.
- **Standardized Communication:** By leveraging OpenBIM standards (BCF), it facilitates communication with external teams on platforms like BIMcollab or BIM Track.
- **Management Visibility:** A decision-making dashboard allows executives to evaluate the project's health at a single glance.



Module 1: Smart Clash Detector (Geometric Synthesis Engine)

This module is the heart of SmartBimPei. It doesn't just find clashes; it understands them to offer concrete solutions.



1.1 The Configuration Panel (Left side)

This panel allows you to configure your search with surgical precision.

- **Model A & Model B Selection:**

 *Usage:* Choose the models to cross-check.

 *Added Value:* Avoids analyzing the entire model unnecessarily.

- **Tolerance (mm):**

 *Usage:* Define the acceptable penetration distance.

 *Added Value:* Prevents "false positives".

- **"Min. Size" Filters:**

 *Usage:* Enter a minimum dimension below which the

element is ignored.

💎 *Added Value*: Drastically reduces calculation time.

- **Auto-Reservation Margin:**

⚙️ *Usage*: Indicates the clearance added around the duct.

💎 *Added Value*: Anticipates insulation and fire stopping.

1.2 The Synthesis Table (Center)

- **Smart Grouping:**

⚙️ *What it does*: Groups close clashes into a single ID.

💎 *Added Value*: Reduces visual fatigue for the coordinator.

- **The "Resa Dimension" Column:**

⚙️ *What it does*: Live display of the final hole dimensions.

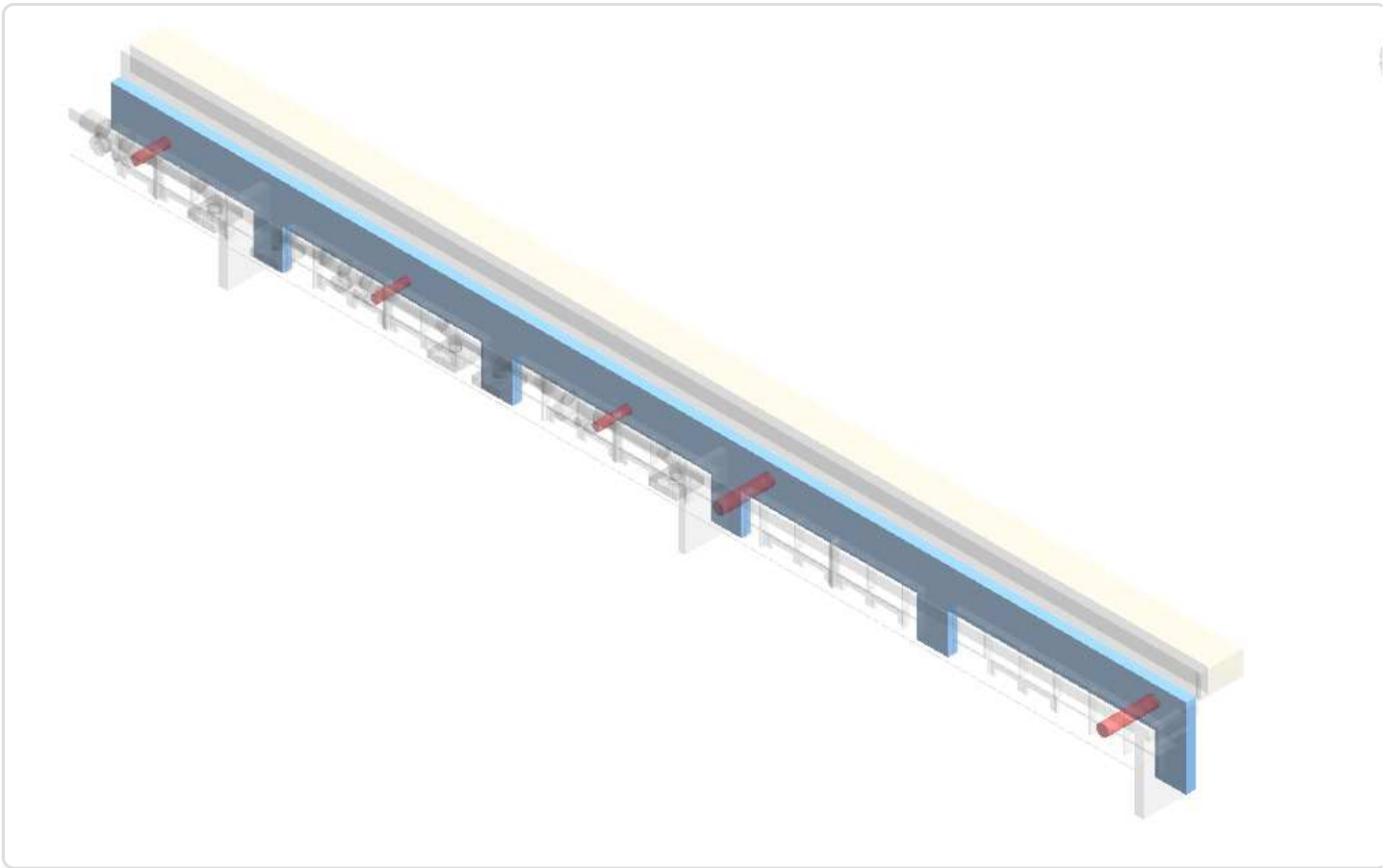
💎 *Added Value*: Architectural impact visible instantly.

1.3 Resolution Actions (Bottom)

👁️ "Isolate Clash" Button:

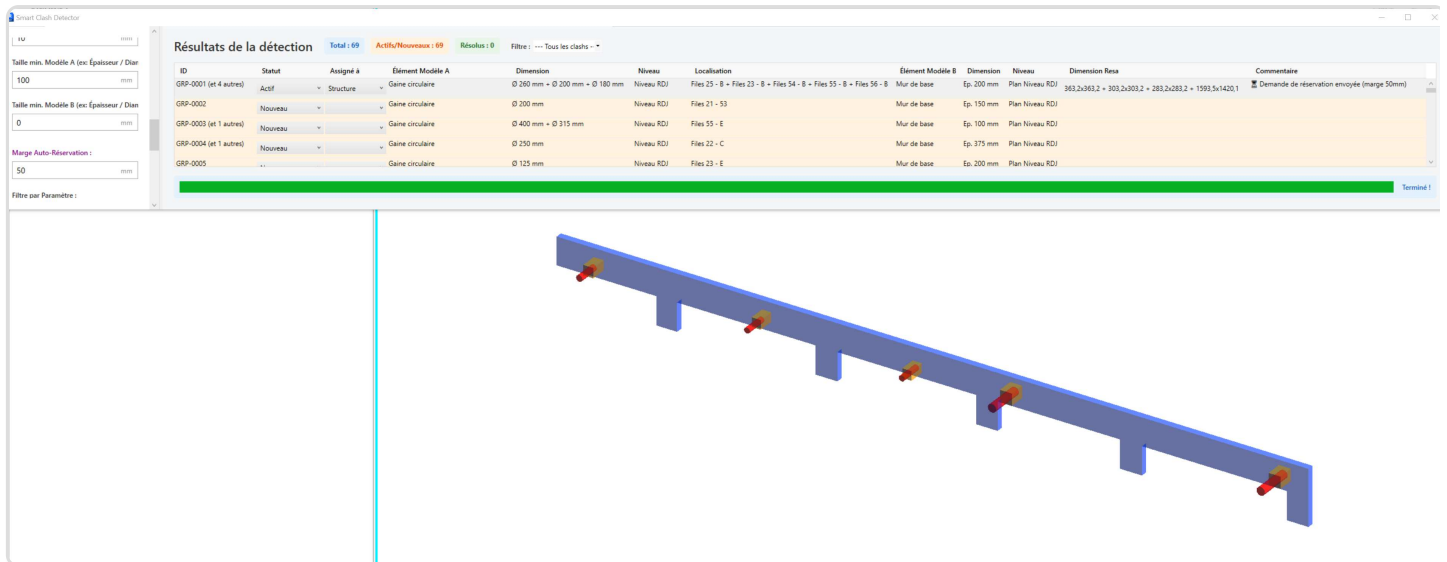
- ⚙️ *Usage*: Opens a centered 3D view.

- 💎 *Added Value*: Bright colors and neutral gray background for 1-second problem understanding.



● "Generate Auto-Reservation(s)" Button:

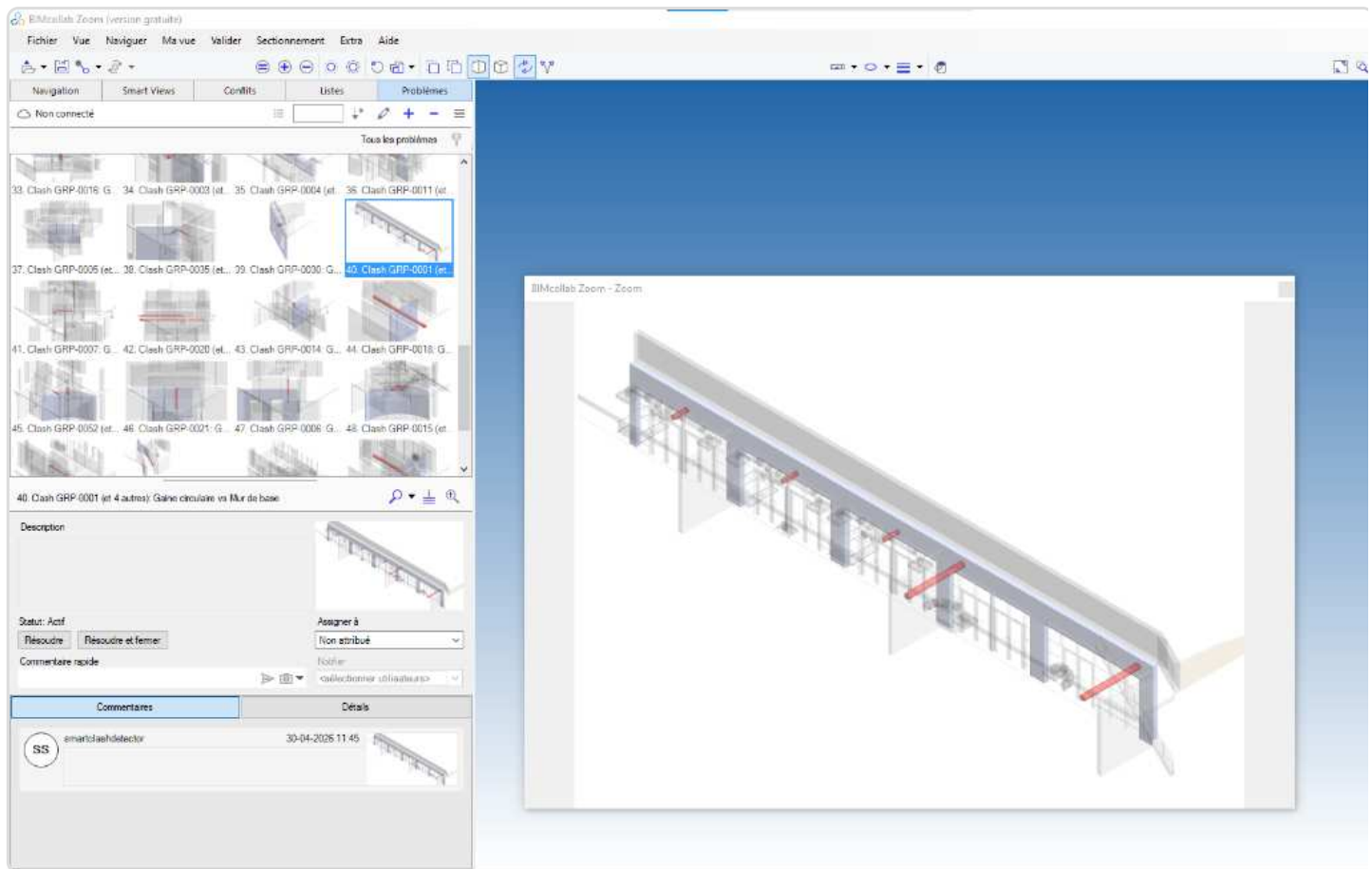
- ⚙️ *Usage:* Places parametric "void" families with a single click.
- 💎 *Added Value:* Manages Sliver Tolerance. What took 10 minutes now takes 1 millisecond.



1.4 Exports & Reports (Communication)

BCF Export (BIM Collaboration Format):

-  *Usage*: Generates a standard .bcfzip file.
-  *Added Value*: Imports into BIMcollab or Navisworks. No more endless email threads.



PDF & CSV Export:

-  *Usage:* Exports a visual report or Excel sheet.
-  *Added Value:* Perfect for coordination meetings.

Rapport de Synthèse - SmartClash

Date : 30/04/2026 11:49

Résumé :

Total Clashes : 55

Actifs/Nouveaux : 55

Résolus : 0

GRP-0001 (et 4 autres) Statut : Nouveau

Assigné à : Non assigné

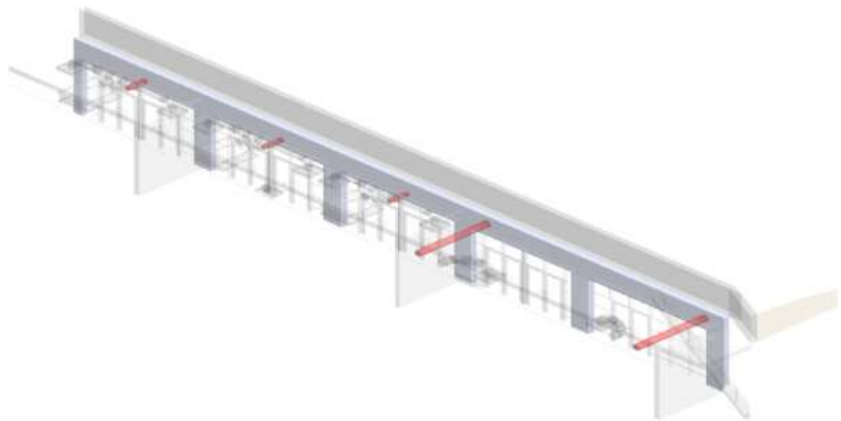
Niveau : Niveau RDJ

Localisation : Aucune grille

A: Gaine circulaire

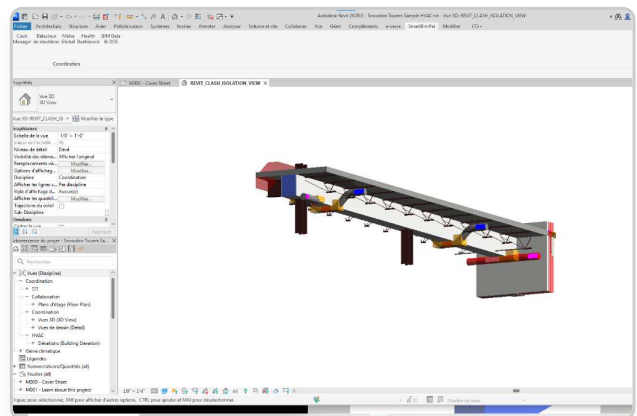
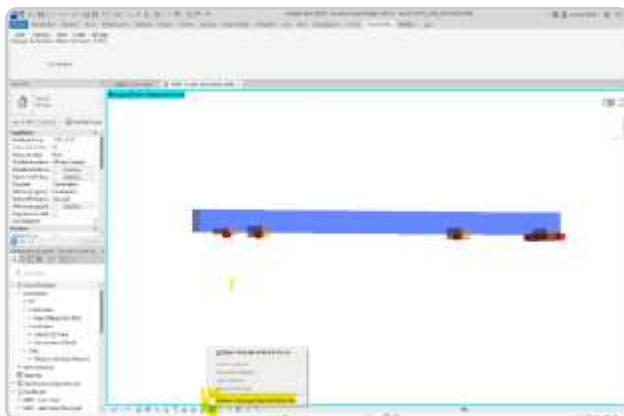
B: Mur de base

Commentaire : Aucun commentaire



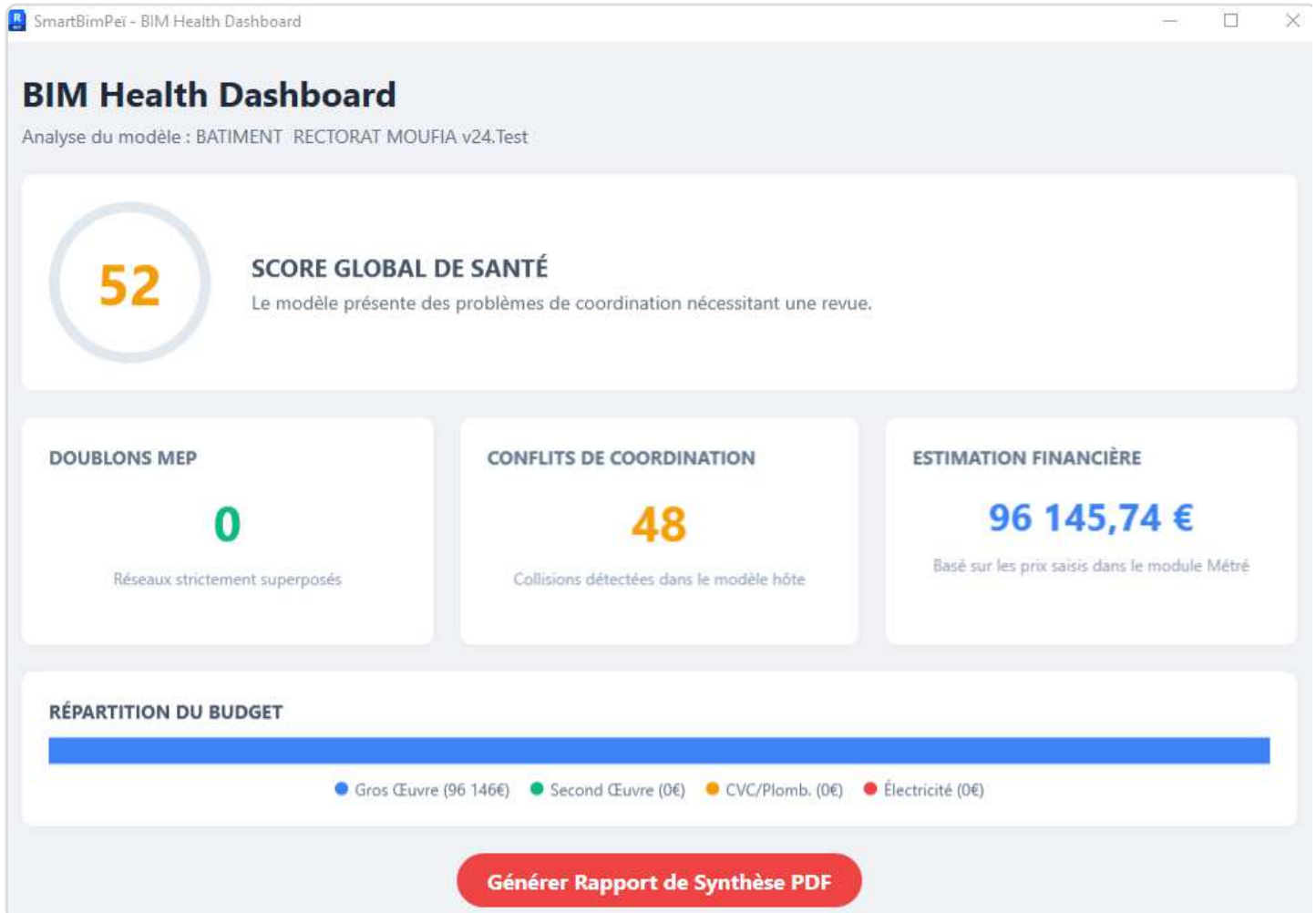
PRO TIP : Visualize the building context

When SmartBimPei generates an isolation view (*REVIT_CLASH_ISOLATION_VIEW*), only the elements involved are visible. To understand exactly where you are in the building model, click on the "**Glasses**" icon at the bottom of the screen and select "**Restore Temporary Hide/Isolate**". The rest of the building will instantly reappear around your clash!





Module 2: BIM Health Dashboard (Management Dashboard)



 **The executive overview:** The Dashboard is the control tower of your project.

-  *Usage:* Analyze key indicators (clashes, duplicates, costs).
-  *Added Value:* Ultimate tool for showcasing accomplished work.



Module 3: Duplicate Detector (Model Cleaner)

Détecteur de doublons

Configuration

Catégories à scanner

- ☐ Matériaux
- ☐ Matériaux de murs-réseau
- ☐ Meubles de rangement
- ☐ Mobilier
- ☐ Modèles de coordination
- ☐ Modèles génériques
- ☒ Murs
- ☐ Usages de points
- ☐ Ossature
- ☐ Couverture de l'usage

Détecter les doublons

Supprimer les copies

Résultats de la détection Total : 7 Actifs/Nouveaux : 7 Résolus : 0

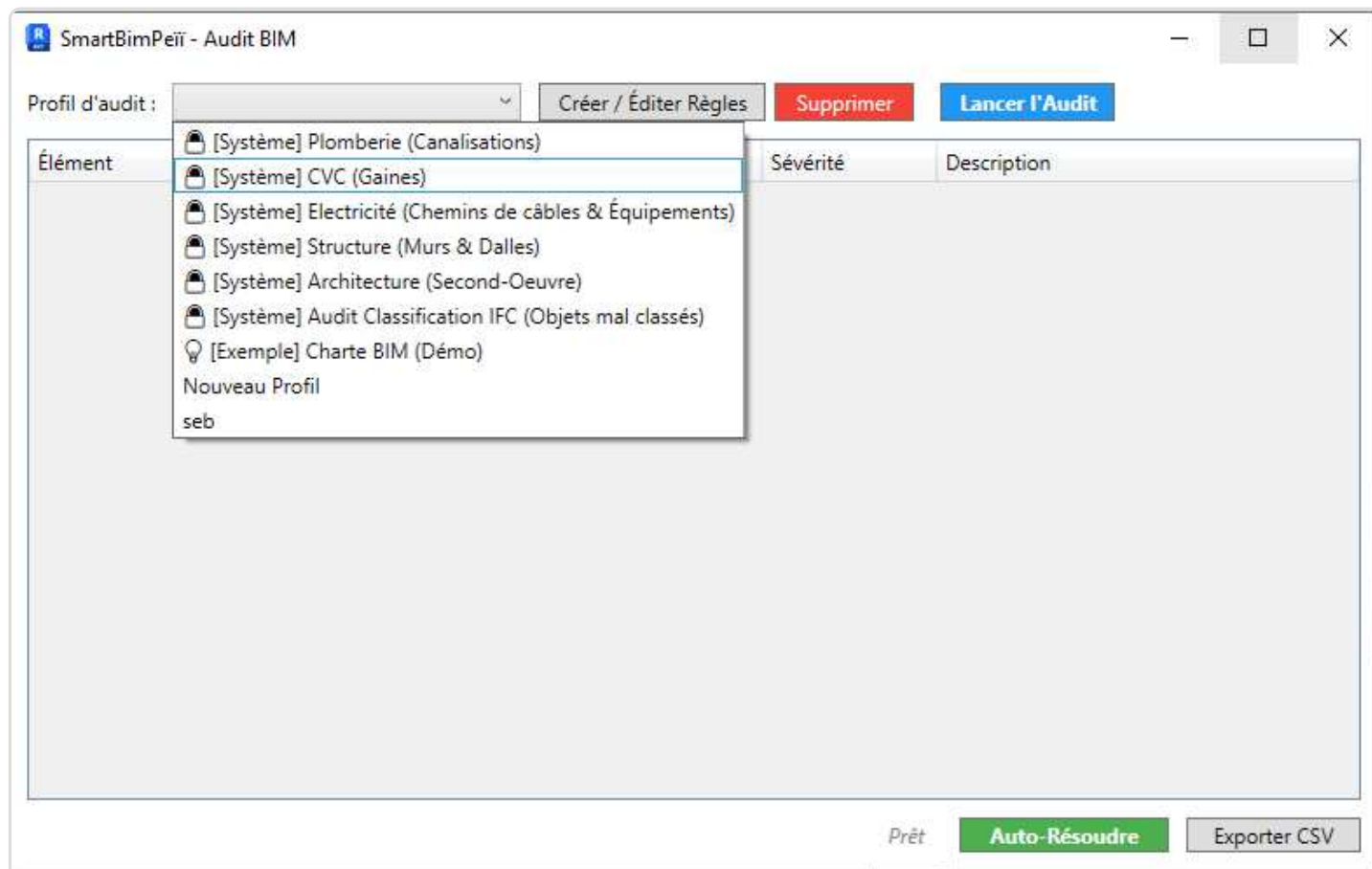
ID	Statut	Élément 1	Maquette 1	Élément 2	Maquette 2	Localisation	Niveau	Dimension	Commentaire
DOUBLON-0001		Mur de base	Snowdon Towers Sample Architectural	Mur de base	Snowdon Towers Sample Architectural	Filés B.1 - Y Axis	R1	Ep. 102 mm	
DOUBLON-0002		Mur de base	Snowdon Towers Sample Architectural	Mur de base	Snowdon Towers Sample Architectural	Filés B.1 - Y Axis	R1	Ep. 102 mm	
DOUBLON-0003		Mur de base	Snowdon Towers Sample Architectural	Mur de base	Snowdon Towers Sample Architectural	Filés B.1 - Y Axis	Block 10 - Parapet	Ep. 102 mm	
DOUBLON-0004		Mur de base	Snowdon Towers Sample Architectural	Mur de base	Snowdon Towers Sample Architectural	Filés X Axis - Y Axis	Block 10 - Parapet	Ep. 102 mm	
DOUBLON-0005		Mur de base	Snowdon Towers Sample Architectural	Mur de base	Snowdon Towers Sample Architectural	Filés X Axis - Y Axis	Block 10 - Parapet	Ep. 102 mm	
DOUBLON-0006		Mur de base	Snowdon Towers Sample Architectural	Mur de base	Snowdon Towers Sample Architectural	Filés C - Y Axis	Block 10 - Parapet	Ep. 102 mm	
DOUBLON-0007		Mur de base	Snowdon Towers Sample Architectural	Mur de base	Snowdon Towers Sample Architectural	Filés C - Y Axis	R2	Ep. 102 mm	

Recherche terminée dans Snowdon Towers Sample Screenshot : 22 : emplacement <Non paginé> (10/10)

- **Usage:** Finds MEP elements stacked in the exact same location.
- **Added Value:** Prevents major financial errors in quantities.



Module 4: BIM Audit (Quality Control & Compliance)



Éditeur de Charte BIM

Nom du Profil : [Exemple] Charte BIM (Démon) Générer depuis TXT (Paramètres Partagés)

Nom de la règle	Catégorie	Type d'audit	Sévérité	Paramètre	Valeur (ou Masque *)	Params Type
Vérifier l'identifiant (Nomma	OST_Walls	Nommage	Avertissement		EXT_*	☒
Présence param. Comment	OST_Doors	PresenceParametre	Bloquant	Commentaires		☒
Statut d'approbation	OST_Windows	ValeurParametre	Avertissement	Statut	Validé	☒

Ajouter une règle
Supprimer la sélection
Mode d'emploi (Guide) ?
Note Nommage : Utilisez * pour ignorer. Ex: PROJET_* ou *_PLAN
Annuler
Sauvegarder la charte

-  *Usage*: Scans the model to verify digital Handover and BIM Execution Plan compliance.
-  *Added Value*: Translates missing data into real impacts on building maintenance.



Module 5: Express Takeoff

(Quantities & Pricing)

SmartBimPei - Mètre Global Structure & MEP

Extraction des Quantités - Maquette Globale (Bêta)

Niveau : Tous Catégorie : Gaine

Niveau	Catégorie	Famille /	Quantité	Surf. (m²) / Long. (m)	Vol. (m³) / Dim.	Prix Unitaire (€)	Total (€)
Niveau R+1	Gaine	Accessoire de canalisation	2	2,65	ø500	0	0.00
Niveau R+1	Gaine	Accessoire de gaine	3	3,47	ø355	10	34.73
Niveau R+1	Gaine	Bouche d'aération	6	19,60	ø200	0	0.00
Niveau R+1	Gaine	Canalisation	19	25,06	ø160	10	250.59
Niveau R+1	Gaine	Equipement de génie climatique	11	22,75	ø315	0	0.00
Niveau R+1	Gaine	Gaine	22	5,28	ø100	0	0.00
Niveau R+1	Gaine	Murs	8	22,41	ø250	10	224.13
Niveau R+1	Gaine	Raccords de canalisation	72	6,14	ø80	10	61.38
Niveau R+1	Gaine	Raccords de gaine	4	2,54	ø125	10	25.36
Niveau R+2	Gaine	Gaine circulaire - France Air_Conduits spir	48	3,74	ø80	10	37.39
Niveau R+2	Gaine	Gaine circulaire - France Air_Conduits spir	6	3,41	ø350	0	0.00
Niveau R+2	Gaine	Gaine circulaire - France Air_Conduits spir	3	6,58	ø160	10	65.80
Niveau R+2	Gaine	Gaine circulaire - France Air_Conduits spir	2	2,61	ø250	10	26.08
Niveau R+2	Gaine	Gaine circulaire - France Air_Conduits spir	6	6,89	ø355	10	68.91
Niveau R+2	Gaine	Gaine circulaire - France Air_Conduits spir	2	2,81	ø330	0	0.00
Niveau R+2	Gaine	Gaine circulaire - France Air_Conduits spir	4	4,03	ø315	0	0.00
Niveau RDC	Gaine	Gaine circulaire - France Air_Conduits spir	4	3,40	ø450	0	0.00
Niveau RDC	Gaine	Gaine circulaire - France Air_Conduits spir	12	29,50	ø315	0	0.00
Niveau RDC	Gaine	Gaine circulaire - France Air_Conduits spir	13	50,49	ø200	0	0.00
Niveau RDC	Gaine	Gaine circulaire - France Air_Conduits spir	3	1,30	ø350	0	0.00
Niveau RDC	Gaine	Gaine circulaire - France Air_Conduits spir	5	1,90	ø260	0	0.00
Niveau RDC	Gaine	Gaine circulaire - France Air_Conduits spir	3	1,16	ø380	0	0.00
Niveau RDC	Gaine	Gaine circulaire - France Air_Conduits spir	4	1,13	ø290	0	0.00

MONTANT TOTAL ESTIMÉ : 3 681,42 €

Scan terminé (8555 éléments reg) [Isoler] [Rétablir] [Actualiser] [Importer (.sbp)] [Exporter (.sbp)] [Exporter Devis (.xlsx)]

- **Usage:** Extracts quantities and cross-references with your pricing library.
- **Added Value:** Colossal time saver for estimators.

*This manual is designed to support the upskilling of teams. **SmartBimPei** is not just a tool; it is a major competitive advantage for your BIM workflows.*

Part 2 : Detailed Interface Guide

This document details the use of the different modules in the SmartBimPei suite.

1. Clash Detection (Clash Detector)

This module is the core of SmartBimPei. It allows you to analyze two models (or a model against itself) to detect physical intersections (clashes) between elements.

1.1 Analysis Configuration

Element	Description and Operation
Model A / Model B	Select the target file (Host Model or Linked Model) for each model. The detection will be run between these two models.
Element Categories	Check the Revit categories to analyze for each model (e.g., Walls, Floors, Ducts, Pipes).
Level (Filter)	Allows restricting the analysis to one or more specific levels of the building to target results.
Sliver Tolerance (mm)	Overlap margin tolerated. If two objects touch over a depth less than this tolerance, the clash is ignored.
Min. Size (Model A & B)	Excludes elements that are too small from the analysis (e.g., ignore pipes with a diameter less than 90mm).
Auto-Provision Margin	Adds a safety margin (in mm) around the clash when creating a provision for void box.
Parameter Filter	Allows precisely targeting certain elements (Exact Parameter Name and Expected Value).
Network Orientation Filter	(Optional) Target only Horizontal or Vertical networks in the detection.

Button: Detect

Starts the detection scan. The results will be displayed in the grid on the right.

1.2 Exports and Render Mode

Element	Description and Operation
Save XML / Load XML	Allows saving the current state of your detection (with resolved/active clashes) and reloading a previous session.
Export BCF / Import BCF	Standard exchange format (BIM Collaboration Format) to send clashes to third-party coordination software (Navisworks, BIMcollab, etc.).
Render Mode (PDF/BCF)	Defines how the software should interpret the clash for the provision for void: - Automatic: Deduces the action according to the nature of the Host Model. - Force Structure Mode: To see/generate the penetrations (actual holes). - Force MEP/Coordination Mode: To see/generate the requests (orange boxes).

Data Flow and Sharing (XML File)

At each detection, SmartBimPei saves your results in an **.xml** file generated automatically in the same folder as your Revit file (.rvt).

To share your analysis with other stakeholders (e.g., Structural Engineer), **always send them the .rvt model AND the .xml file in the same folder**. They will then see the entire clash history directly in their interface!

2. Module 4: BIM Audit (Unified Version)

This module now centralizes all quality control for your model. It merges the former data verification tool with the Audit system, allowing you to instantly validate compliance with your BIM Execution Plan (BEP) or shared parameters.

2.1 Main Window (Running the Audit)

Control	Description
Audit Profile (Dropdown)	Select a pre-configured system profile or choose a custom profile (BIM Charter) you have created.
Create / Edit Rules	Opens the BIM Charter Editor to modify the selected profile or create a new one.
Run Audit	Scans the entire Revit model according to the chosen profile's rules and displays the list of non-conformities (Element, Severity, Description).
Auto-Resolve	Attempts to automatically fix certain common errors selected in the list.
Export CSV	Exports the audit results to an Excel-compatible CSV file for your coordination reports.

2.2 BIM Charter Editor (Creating Rules)

This is where you define the strict data rules for your project.

Control	Description
Generate from TXT (Shared Parameters)	<i>Key Feature.</i> Directly import the shared parameters .txt file provided by the BIM Manager. The plugin will automatically generate all the necessary rules to check if these parameters are present!
Add a rule	Adds a blank row to create a manual rule. You can check naming conventions (using * wildcard), parameter presence, or exact parameter values.
Delete Selection	Removes the currently selected rule in the grid.

User Guide

Displays quick help on how to use input masks.

Save Charter

Saves your custom rules under the profile name entered at the top left.

3. Duplicate Detector

This module scans the model to quickly identify overlapping elements (modeled twice in the same place).

Element	Description and Operation
Categories to scan	Check the object categories where you suspect duplicates (e.g., Walls, Plumbing fixtures, etc.). Very useful for checking imports or copy-paste errors.
Blue button: Detect duplicates	Analyzes the selected categories and displays the list of perfectly overlapping elements in the results table (with their ID and location).
Red button: Delete copies	Automatically deletes one of the two duplicate elements to clean up your model. Important note: This button will only delete duplicates present in your Host Model. If the duplicate comes from a Linked Model (IFC or RVT), it will be ignored because Revit prohibits modifying links (read-only).

4. Quantity Extraction (Global Takeoff)

This module generates precise quantities (lengths, areas, volumes) from your model to prepare your estimates and costing.

Element	Description and Operation
Filters (Model, Level, Category)	These drop-down menus at the top become active after extraction. They allow you to quickly filter the table display (e.g., only show Level 1).
Red button: Run Extraction	Scans the entire model (and its links) to extract the complete list of elements with their actual physical quantities.

Blue buttons: Isolate / Restore	Select a row in the table and click 'Isolate' to only show this object in 3D in Revit. 'Restore' cancels this isolation.
Green button: Export Estimate (.xlsx)	Generates a perfectly formatted Excel file containing all the data from your table, ideal for costing.
Import/Export buttons (.sbp)	Allows saving your current table (and especially your entered unit prices) to a proprietary .sbp file to reload it later.
Total Estimated Amount	Dynamically updates at the bottom right based on the unit prices you entered in the table.

5. Dashboard (Global Dashboard)

The control center that monitors the health of your projects and the progress of your BIM coordination.

Element	Description and Operation
Progress charts	Visually displays the ratio of "Resolved" vs "Active" clashes by reading the data from your XML file.
Health Indicator (Score)	Overall score given to your model based on data quality (parameters) and the presence of duplicates.
Button: Refresh data	Forces a reread of the XML file and the model to update all your charts in real time.